AnaTele

Evidence-first investigation of ArduPilot flight logs
Architecture, mathematics and design decisions

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Abstract

AnaTele is an open-source desktop tool that reads ArduPilot flight recordings (Mission Planner `.tlog`, DataFlash `.bin`/`.log`, mapped `.csv`) and turns them into an investigation: when the aircraft actually took off, what each subsystem was doing before the event, which screening rules fire, and how the plausible causes rank against each other. Every statement is linked to the exact log record that supports it. This report documents the motivation, the system architecture, and all of the mathematics used: binary framing, clock segmentation, signal normalisation, flight-phase inference, thrust-vector tracking error, motor saturation and mixer decomposition, persistence and counter detectors, the hypothesis-ranking score, waypoint geometry, plot down-sampling, 3D replay kinematics, the fault-injection flight simulator, and the integrity model of the shareable `.ant` analysis format.

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1 Motivation

AnaTele started with a crash. While leading the NIDAR *RescueSwarm* problem statement, our team began offline flight testing with cameras and our own algorithms, and the aircraft went down twice. Afterwards the investigation looked like this:

1. Thousands of lines of telemetry, read by hand.
2. A general-purpose AI assistant, which correctly pointed at the compass (it had not been calibrated properly, which is now part of our pre-flight checklist) but could say nothing else about the flight.
3. Mission Planner's graphing, which was slow on long logs and whose heading display was confusing exactly when the yaw estimate was wrong.

Reading the log manually showed that there was much more in it: the order of events before the disarm, how the motors were being driven, whether vibration was a cause or only a consequence of the impact, and which questions the log simply could not answer. AnaTele automates that reading. It works out the flight phases, evaluates a versioned rule pack, ranks candidate causes using the evidence for and against each one, plots the signals that matter, and replays where the aircraft actually went.

1.1 Design principles

These rules come from the design document (`docs/DESIGN.md`) and apply everywhere in the code:

Observation ≠ interpretation. Decoded records are immutable evidence. Derived signals, findings and hypotheses are labelled as such and point back to record ids.

Missing is unknown, never healthy. Every rule result is one of four states: *triggered*, *not triggered*, *unassessable*, *not applicable*. Absent data produces *unassessable*, not a pass.

No invented probabilities. Hypotheses are ordered by an explicit, inspectable screening score, never by a probability (Section 9 explains why).

Read-only. There is no live MAVLink link, no parameter write and no flight command.

Reproducible. Engine, rule-pack and semantic-pack versions and SHA-256 hashes are stored with every result.

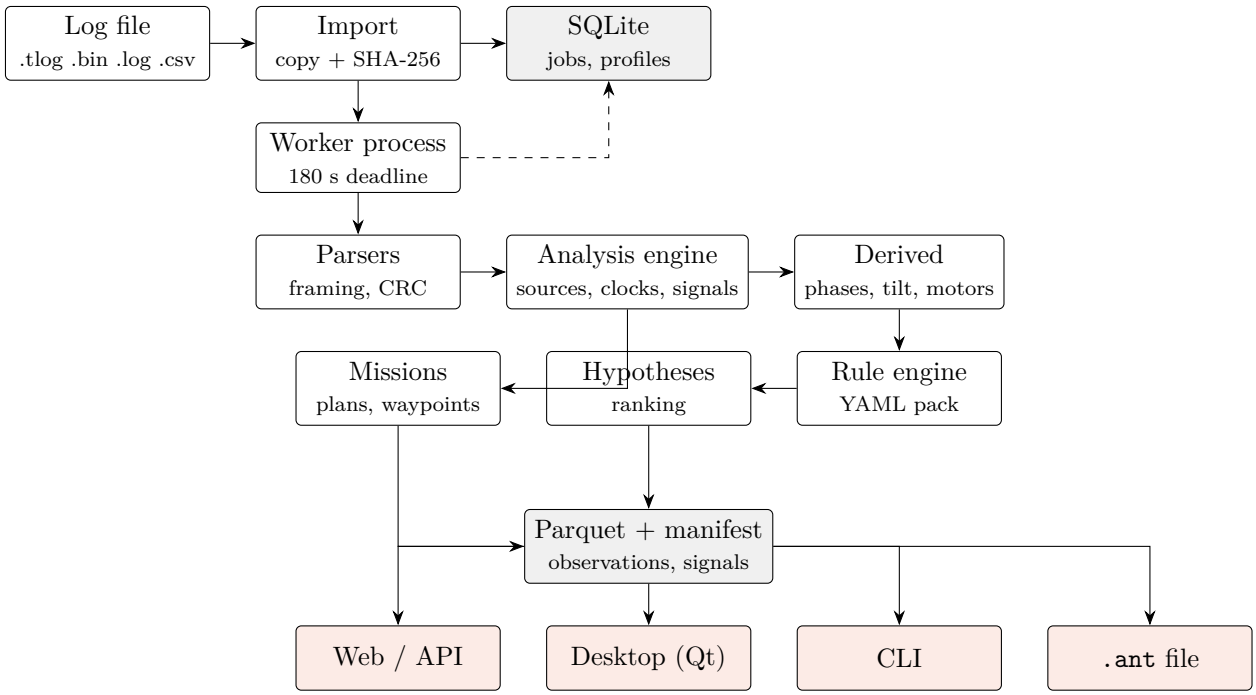


Figure 1: Data flow. Shaded boxes hold persistent state; orange boxes are user interfaces.

2 System architecture

2.1 Components

Module	Responsibility
`parsers.py`	File-only readers. Timestamped MAVLink framing with CRC validation and resync; DataFlash via pymavlink `DFReader`; a strict CSV schema. Each record gets an id, byte offset and length.
`analysis.py`	Streaming engine: source classification, clock epochs, arming episodes, STATUSTEXT reassembly, events, parameter snapshots, signal extraction. Writes Parquet and the manifest.
`derived.py`	In-memory time-series store, flight-phase inference, tilt tracking error, motor mapping and saturation flags, mixer decomposition, window statistics.
`rules.py`, `rules/core.yaml`	Declarative rule pack (20 rules, 9 detector types) and hypothesis ranking. YAML is loaded with `safe_load`; rules cannot execute code.
`semantics.py`, `semantics/*.yaml`	Firmware-specific meaning (ArduCopter 4.7.2 tied to its git hash), parameter decoding, volatile parameters.
`missions.py`	Mission upload/download transactions, plan versions, waypoint status.
`compare.py`	Parameter diff, phase-aligned flight comparison, per-aircraft trends.
`store.py`, `worker.py`	SQLite job queue, data directories, worker process with a deadline.
`desktop.py`, `api.py`, `cli.py`	PySide6 + matplotlib desktop app, FastAPI web UI, command line.
`replay.py`	3D flight path and attitude glyph, PNG/GIF export.
`simulator.py`	Physics-based quadcopter with fault scenarios that writes real `.tlog` files.
`antfile.py`	Shareable `.ant` analysis container: export, validation, import.

2.2 Process and storage model

Importing copies the file into `uploads/`, computes its SHA-256 and inserts a *queued* job row. A scheduler claims jobs with an `BEGIN IMMEDIATE` transaction (so two schedulers can never claim the same job) and starts a **separate worker process**. A parser crash, a memory blow-up or a pathological file can therefore not take down the UI, and the worker is killed after 180 s. The worker streams records into two CSV spools, which are converted to columnar **Parquet** files:

- `observations.parquet`: one row per decoded record (time, clock, source, epoch, message, offset, length, sequence, quality, JSON payload). This is the evidence store.
- `signals.parquet`: normalised long-format samples $(id, t, epoch, source, name, value, unit)$ including derived signals.

The manifest (findings, rule results, phases, missions, metrics, limitations, hashes) is stored as JSON in the job row. Queries over Parquet use an embedded DuckDB with a 256 MB memory limit and two threads.

2.3 Resource limits

Upload 100 MiB; 10^6 records; 20 000 events; 6×10^6 in-memory derived samples; 64 KiB maximum resync span; `.ant` file 1 GiB. Each limit fails loudly; none silently truncates evidence.

3 Ingestion mathematics

3.1 Telemetry log framing

A Mission Planner `.tlog` is a concatenation of

$$\underbrace{t_{\mathrm{UTC}}}_{\text{8 bytes, big-endian, } \mu s} \quad \| \text{ MAVLink frame.}$$

For a frame with start marker m and payload length n, the header length is $h = 10$ for MAVLink 2 ($m =$ `0xFD`) and $h = 6$ for MAVLink 1 ($m =$ `0xFE`). With signing flag $s \in \{0, 1\}$ the total frame length is

$$L = h + n + 2 + 13s,$$

where 2 is the CRC and 13 the signature. A candidate is accepted only if

$$m \in \{\texttt{0xFD}, \texttt{0xFE}\} \quad \wedge \quad 946\,684\,800 \times 10^6 \le t_{\mathrm{UTC}} \le 4\,102\,444\,800 \times 10^6,$$

i.e. the timestamp lies between 2000-01-01 and 2100-01-01, and if the CRC-16/MCRF4XX (X.25) over the header and payload, seeded with the message's `CRC_EXTRA`, matches. On failure the reader advances one byte and retries (*resync*). If more than 65 536 bytes pass without a valid frame, recovery stops and the file is rejected rather than guessed at. Every resynchronised span, truncated tail, unknown message id and unsupported incompatibility flag becomes a diagnostic with a byte offset. Signed frames are kept and labelled *signature unverified*.

3.2 Clock epochs

A log mixes clocks: the ground station's capture clock (UTC), the autopilot's `time_boot_ms`, and DataFlash `TimeUS`. Reboots and capture glitches make them jump backwards, and comparing samples across such a jump would be meaningless. AnaTele therefore assigns each record an *epoch*

$$e = 10\,000\,c + b,$$

where c counts capture-clock regressions and b counts per-source boot regressions:

$$\begin{aligned}
&c \leftarrow c + 1 && \text{if } t_{\mathrm{capture}} < t_{\mathrm{capture}}^{\mathrm{prev}} - 2 \text{ s},\\
&b \leftarrow b + 1 && \text{if boot_ms} < \text{boot_ms}^{\mathrm{prev}} - 10\,000 \text{ and not a wrap},\\
&b \leftarrow b + 1 && \text{if TimeUS} < \text{TimeUS}^{\mathrm{prev}} - 10^6 \text{ (DataFlash)}.
\end{aligned}$$

A uint32 wrap is recognised when the previous value exceeds `0xF0000000` and the new one is below 10^5. All series, episodes and rules are keyed by *(source, epoch)*; an open armed episode crossing an epoch boundary is closed with the note "clock discontinuity; end time unknown".

3.3 Source classification

Each MAVLink `(system, component)` pair is a source. Its role comes from what it says, not from its id:

- a `HEARTBEAT` with `type = MAV_TYPE_GCS` ⇒ ground station;
- a `HEARTBEAT` with a valid autopilot ⇒ vehicle;

- `RADIO_STATUS` from an unclassified source ⇒ radio.

Only vehicle sources may arm. This prevents the classic mistake of reading the GCS heartbeat's `base_mode` as the aircraft's arming state. Armed state is

$$\text{armed} = (\text{base_mode} \,\&\, \texttt{MAV_MODE_FLAG_SAFETY_ARMED}) \neq 0,$$

and each armed interval becomes an *episode*.

3.4 Signal normalisation

Raw fields are converted to SI-like units with explicit scale factors, for example:

$$\varphi_{\text{deg}} = \tfrac{180}{\pi}\varphi_{\text{rad}}, \quad \text{lat}_{\text{deg}} = 10^{-7}\,\text{lat}_{\text{int}}, \quad h_{\text{m}} = 10^{-3}\,h_{\text{mm}}, \quad V = 10^{-3}\,V_{\text{mV}}, \quad I = 10^{-2}\,I_{\text{cA}}.$$

Sentinels are treated as absence, not as measurements: a battery voltage of 0 or 65 535 (no monitor configured), current -1, RC values 0 or 65 535, and non-finite floats are dropped.

3.5 Parameter decoding

`PARAM_VALUE` transports every parameter as a 32-bit float, but integer parameters may be encoded either by *C-cast* (float(v)) or *bytewise* (the integer's bytes reinterpreted as a float). AnaTele uses the vehicle's `AUTOPILOT_VERSION` capability bits, or the firmware semantic pack, to choose. Under C-cast a value is accepted only if it is integral and inside the declared type's range:

$$\text{decode}(r,\tau) = \begin{cases} \lfloor r \rfloor & r \in \mathbb{Z},\ r \in [\ell_\tau, u_\tau] \\ \text{unverified} & \text{otherwise} \end{cases} \qquad [\ell_\tau, u_\tau] \in \{[0, 2^8\text{–}1], [-2^7, 2^7\text{–}1], \ldots, [-2^{31}, 2^{31}\text{–}1]\}.$$

Under bytewise encoding the original 4 wire bytes (kept in the record) are unpacked little-endian with the declared type. If both capability bits are set, the encoding is *ambiguous* and the value stays unverified, so rules that depend on it become *unassessable*.

Snapshot completeness compares the observed indices $\mathcal{I}$ with the reported count N: complete iff $|\{0,\ldots,N-1\} \cap \mathcal{I}| = N$. Even then it is labelled "not atomic" because values arrive at different times.

4 Time-series store

Each ($source, epoch, name$) owns three parallel typed arrays t_i (int64 μs), v_i (float64) and r_i (record id), kept sorted by t. Out-of-order samples are inserted with binary search. Range queries return the index interval

$$[\,\text{lo}, \text{hi}) = [\,\text{bisect_left}(t, a),\ \text{bisect_right}(t, b)\,)$$

in $O(\log n)$. The *nearest-sample* query used to pair signals is

$$\text{nearest}(t^*, \delta) = \arg\min_{i \in \{k-1,k\}} |t_i - t^*| \quad \text{subject to } |t_i - t^*| \le \delta, \qquad k = \text{bisect_left}(t, t^*),$$

returning nothing if no sample lies within δ. AnaTele never interpolates evidence. Pairing uses the nearest real sample or nothing.

5 Flight-phase inference

Rules must distinguish "before the problem" from "during the crash": vibration at impact is a consequence, not a cause. Phases are therefore computed per armed episode $[t_a, t_d]$.

Reported. If the vehicle sends `EXTENDED_SYS_STATE` with landed state *In air* or *Taking off*, the first such time is the takeoff (status = *airborne_reported*).

Inferred. Otherwise, with home-relative altitude samples h_i:

$$\begin{aligned}
\bar{h} &= \text{median}\{h_i : t_a \le t_i \le t_a + 2\text{ s}\} && \text{(armed baseline)}\\
i_{\text{rise}} &= \min\{i : h_i - \bar{h} \ge 1.0\text{ m}\} && \text{(clear climb)}\\
\text{demand} &: \exists\, j :\ t_j \in [t_{\text{rise}} - 5\text{ s}, t_{\text{rise}}],\ \text{throttle}_j \ge 10\%\ \vee\ u_{k,j} \ge 0.25 && \text{(the motors were driving it)}\\
i_{\text{takeoff}} &= \max\{i < i_{\text{rise}} : h_i - \bar{h} \le 0.2\text{ m}\} && \text{(backtrack to lift-off)}
\end{aligned}$$

where $u_{k,j}$ is motor k's normalised output (Section 7). No rise gives *no_airborne_evidence*. A rise without demand is treated as estimator drift. No altitude at all gives *unassessable*. The median baseline makes the estimate robust to a few noisy barometer samples at arming.

Event markers.

$$\begin{aligned}
t_{\text{excursion}} &= \min\{t_i \ge t_{\text{takeoff}} : |\varphi_i| > 60^\circ \vee |\theta_i| > 60^\circ\},\\
t_{\text{pre-end}} &= \min\big(t_{\text{excursion}},\ t_{\text{crash}} - 3\text{ s}\big),\\
t_{\text{air-end}} &= \min\big(t_{\text{excursion}},\ t_{\text{crash}} - 3\text{ s},\ \max\{t_i : h_i - \bar{h} \ge 0.5\},\ t_d\big).
\end{aligned}$$

The 3 s guard keeps the impact out of the *pre-event* window. Rules then choose one of five windows: *airborne*, *airborne_pre_event* $= [t_{\text{takeoff}}, t_{\text{pre-end}}]$, *airborne_with_event*, *event_and_after* and *armed_pre_takeoff*. All thresholds are recorded in the output, because they are analysis choices, not firmware behaviour.

6 Thrust-vector (tilt) tracking error

A multirotor can only push along its body z axis, so the meaningful control error is the angle between where the controller wanted the thrust to point and where it actually pointed. For roll φ and pitch θ (ZYX Euler, yaw ignored because it does not tilt the thrust), the body thrust axis expressed in a yaw-free level frame is

$$\hat{\mathbf{z}}(\varphi, \theta) = \begin{pmatrix} \sin\theta\cos\varphi \\ -\sin\varphi \\ \cos\theta\cos\varphi \end{pmatrix}, \qquad \|\hat{\mathbf{z}}\| = 1.$$

The tracking error between desired (φ_d, θ_d) and estimated (φ, θ) is the great-circle angle

$$\boxed{\varepsilon = \arccos\big(\text{clip}_{[-1,1]}\ \hat{\mathbf{z}}(\varphi_d, \theta_d) \cdot \hat{\mathbf{z}}(\varphi, \theta)\big)}$$

Unlike $|\varphi_d - \varphi| + |\theta_d - \theta|$, this is frame-independent and does not double-count combined roll and pitch errors. ArduPilot's own crash check uses the same idea (an angle error above 30° for 2 s).

Sources of the desired attitude. `NAV_CONTROLLER_OUTPUT` (`nav_roll`, `nav_pitch`), `ATTITUDE_TARGET` or DataFlash `ATT.DesRoll`. The first one seen is locked per source so that two definitions are never mixed. A quaternion target $q = (w, x, y, z)$ is converted with

$$\varphi = \text{atan2}\big(2(wx + yz),\ 1 - 2(x^2 + y^2)\big), \qquad \theta = \arcsin\big(\text{clip}(2(wy - zx))\big).$$

Pairing. For each desired-roll sample at t, desired pitch must lie within 1 ms and estimated roll/pitch within 250 ms (`nearest`). Otherwise no error sample is produced. The limitation that desired and estimated values are sampled at different instants is reported in the output.

7 Motors: saturation and mixer decomposition

7.1 Mapping

Motors are found from `SERVOn_FUNCTION` $\in \{33, \ldots, 40\}$ (motor $1 \ldots 8$). With PWM range $[P_{\min}, P_{\max}]$ (`MOT_PWM_*`, falling back to `RC3_*`) and spin limits $s_{\min}, s_{\max}$ (`MOT_SPIN_*`):

$$\Delta P = P_{\max} - P_{\min}, \qquad P_{\text{floor}} = P_{\min} + s_{\min}\Delta P, \qquad P_{\text{ceil}} = P_{\min} + s_{\max}\Delta P.$$

For each output sample P_k the normalised command and saturation flags are

$$u_k = \frac{P_k - P_{\min}}{\Delta P}, \qquad F_k^{\text{floor}} = [P_k \le P_{\text{floor}} + 2], \qquad F_k^{\text{ceil}} = [P_k \ge P_{\text{ceil}} - 2],$$

with a $\pm 2\,\mu$s tolerance for PWM quantisation. Samples where all outputs are at or below $P_{\min} + 1$ (disarmed or stopped) are skipped so that idle is not mistaken for saturation. If any of these parameters is unknown, the motor rules are *unassessable* and say why.

7.2 Mixer decomposition

Each motor k has a geometry vector (x_k, y_k, σ_k): x = right (± 1), y = forward (± 1), $\sigma = +1$ for counter-clockwise props. For QUAD/X:

$$1 : (+1, +1, +1), \quad 2 : (-1, -1, +1), \quad 3 : (-1, +1, -1), \quad 4 : (+1, -1, -1).$$

The share of the commands that goes into each axis is the normalised weighted sum

$$D_{\text{roll}} = \frac{\sum_k x_k c_k}{\sum_k |x_k|}, \qquad D_{\text{pitch}} = \frac{\sum_k y_k c_k}{\sum_k |y_k|}, \qquad D_{\text{yaw}} = \frac{\sum_k \sigma_k c_k}{\sum_k |\sigma_k|},$$

where c_k is the PWM (time series) or the mean $\bar{u}_k$ (rule). $D_{\text{roll}} > 0$ means the right side works harder (the frame is being held up on the right), and $D_{\text{yaw}} > 0$ means the CCW propellers work harder.

7.3 Imbalance rule

Over the pre-event window (first 2 s trimmed to exclude the spool-up),

$$\bar{u}_k = \frac{1}{n_k} \sum_i u_{k,i}, \qquad \text{spread} = \max_k \bar{u}_k - \min_k \bar{u}_k, \qquad \text{triggered} \iff \text{spread} > 0.12.$$

If $|D_{\text{yaw}}| > 2\max(|D_{\text{roll}}|, |D_{\text{pitch}}|)$, the report says that the imbalance is predominantly yaw (one prop direction working harder). This points to twisted mounts, a prop or direction mismatch, or sustained yaw demand, rather than a single weak motor. A single weak motor shows up instead as roll and pitch imbalance towards that corner.

8 Rule engine

Rules are YAML data with an id, an integer version, a severity, a scope (*event, source, episode, series, snapshot*), a detector type and text templates. The whole pack is hashed with SHA-256. The detectors are below.

8.1 Persistence (vibration, tracking error, EKF ratio, saturation)

Given samples (t_i, v_i) in the windows W, a threshold L, a maximum gap g and a duration T, a *run* is a maximal sequence of consecutive samples with

$$v_i > L \quad \text{and} \quad t_i - t_{i-1} \le g.$$

Its duration is $d = t_{\text{last}} - t_{\text{first}}$. The rule triggers if

$$\max_{\text{runs}} d \;\ge\; T.$$

A gap larger than g breaks the run, so a dropout cannot be counted as "still above". The core criteria are as follows.

Rule	Signal	Criterion
vibration-persistent-pre-event	`imu.*.vibration.*`	> 30 m/s^2 for 2 s, or > 60 for 1 s
tracking-loss	tilt error ε	$> 30°$ for 0.5 s
ekf-compass-innovation-pre-event	`ekf.*compass_ratio`	> 1.0 for 2 s
motor-ceiling-saturation	F_k^{ceil}	> 0.5 for 1 s
motor-floor-saturation-pre-event	F_k^{floor}	> 0.5 for 2 s

The vibration threshold can come from the aircraft profile. An EKF test ratio above 1 means that the innovation exceeded its gate (the measurement disagreed with the prediction by more than the filter's tolerance).

Resolution check (Nyquist-style). If no run qualifies, absence is only informative when the sampling can resolve the criterion. With the median sample interval $\widetilde{\Delta t}$,

$$\widetilde{\Delta t} \;\ge\; T \text{ for every criterion} \implies \text{unassessable},$$

because a 2 s event cannot be ruled out by 2 s-spaced samples. If only some criteria are unresolvable, the result is *not triggered* and the text names the unresolved ones. Fewer than three samples in the window also gives *unassessable*.

8.2 Counter increments (accelerometer clipping)

Clip counters are cumulative. For consecutive pairs with $t_{i+1} - t_i \le 3$ s,

$$\Delta_i = v_{i+1} - v_i, \qquad \text{added} = \sum_i \max(0, \Delta_i), \qquad \text{resets} = |\{i : \Delta_i < 0\}|.$$

Negative steps (reboot or reset) are excluded rather than subtracted. The increment is reported separately for the pre-event window and the event and after, so "clipping at impact" is not confused with "clipping in flight".

8.3 Event presence and sequence

Presence matches event kinds or a regular expression on reassembled STATUSTEXT. If the source never recorded any text, the result is *unassessable*, not *not triggered*. Optional context signals report the range within $\pm w$ seconds and compare it with a parameter (for example, mapped throttle versus `FS_THR_VALUE`).

Sequence looks for a precursor p before an anchor a (crash-check disarm):

$$0 \le t_a - t_p \le 20 \text{ s}.$$

It then adds *support* if the EKF compass ratio was > 1 for ≥ 2 s in the 30 s before p, and a *contradiction* if tilt error was already $> 30°$ for ≥ 0.5 s before p (the aircraft was already out of control, so the compass switch is more likely recovery than cause).

8.4 Peak, coverage and parameter detectors

Peak: $\max_i v_i > L$ with the argmax record as evidence. *Coverage*: triggers when no battery signal exists during any armed episode ("power cannot be assessed"). *Parameter*: equality or bit test $\lfloor v \rfloor \,\&\, 2^b \neq 0$, only for verified values and, for bit meanings, only under the matching firmware semantics. *Profile*: $|v_{\text{obs}} - v_{\text{exp}}| > 10^{-4}$ for each declared expected parameter.

9 Hypothesis ranking

For each episode with a crash-check report, every hypothesis H (estimator/magnetic, thrust/configuration, vibration, power) declares supporting rules r, each with an evidence *family* $f(r)$ and a weight w_r, plus contradicting conditions with penalties π_c.

$$S(H) = \sum_f \min\Big(C, \sum_{\substack{r \in H,\ f(r)=f \\ r \text{ triggered}}} w_r\Big) - \sum_{c \in H,\ c \text{ holds}} \pi_c, \qquad C = 1.5.$$

The family cap C stops correlated evidence from inflating the score. Ten reports from the same saturated motor are one piece of propulsion evidence, not ten. The labels are as follows.

Condition	Label
blocked (e.g. no power monitoring), or nothing assessed and nothing against	unassessable
support and contradictions both present	mixed support
$S \geq 2$	strong support
$1 \leq S < 2$	some support
$0 < S < 1$	weak support
otherwise	limited or contradicting evidence

Hypotheses are sorted by (unassessable last, $-S$). For example, thrust/configuration:

$$S = \min(1.5,\, 1_{\text{tracking}}) + \min(1.5,\, 1_{\text{floor}} + 0.75_{\text{imbalance}} + 0.5_{\text{ceiling}}).$$

With tracking loss, ceiling saturation and imbalance this gives $1 + 1.25 = 2.25$, which is *strong support*.

Why not a probability? A posterior $P(H \mid E)$ needs priors (base rates of each failure for this airframe) and likelihoods $P(E \mid H)$ that account for strong dependence between signals. Neither exists for a single hobby or competition aircraft. Printing "73 %" would imply calibration that the data cannot support. The score is an expert-weighted *screening order*: transparent, reproducible, and shown together with what supports it, what contradicts it, and which measurement would discriminate ("*ESC RPM telemetry and a motor-order test*").

10 Missions and waypoint geometry

10.1 Transactions and plan versions

Upload (`MISSION_COUNT`→`REQUEST`→`ITEM`→`ACK`), download, partial write and clear are reconstructed per (GCS, vehicle, mission type, epoch). Retransmissions, restarts, conflicting items and broadcast ACKs are handled. Only an accepted transfer in which every item was observed creates a plan version with high confidence. Otherwise the plan is marked uncertain.

10.2 Distance

The horizontal distance between a recorded position and a waypoint uses the haversine formula with the IUGG mean Earth radius $R = 6\,371\,008.8\,\mathrm{m}$:

$$a = \sin^2\tfrac{\Delta\phi}{2} + \cos\phi_1 \cos\phi_2 \sin^2\tfrac{\Delta\lambda}{2}, \qquad d = 2R \arcsin\!\big(\min(1, \sqrt{a})\big).$$

10.3 Waypoint status

For each item, let $d^* = \min_i d(p_i, w)$ over the episode's recorded positions, r the acceptance radius (`WP_RADIUS_M`, or `WPNAV_RADIUS`/100), and σ_h the GPS horizontal accuracy at that instant:

$$\text{status} = \begin{cases} \text{reported_complete} & \texttt{MISSION_ITEM_REACHED}(seq) \text{ observed} \\ \text{geometry_consistent_with_completion} & d^* \le r + \sigma_h \\ \text{not_visited_in_observed_trajectory} & d^* > r + \sigma_h \ \wedge\ \max \Delta t_{\text{pos}} \le 5\ \text{s} \\ \text{completion_not_observed} & \text{otherwise (coverage gaps)} \\ \text{unassessable} & \text{no position or no target} \end{cases}$$

The coverage condition matters: "not visited" is only claimed when there is no gap longer than 5 s in which the aircraft could have passed the waypoint unrecorded. No interpolation is performed.

11 Statistics, comparison and plotting

Window statistics. For values $v_{(1)} \le \cdots \le v_{(n)}$ in a window: n, mean, min, max and $p_{95} = v_{(\min(n, \lfloor 0.95n \rfloor + 1))}$ (nearest-rank). These feed the Flight Statistics grid, the comparison table and the trends.

Comparison and trends. Two flights are aligned by *phase*, not by clock: metrics are computed over each one's own pre-event airborne window. Parameter diffs ignore volatile parameters and use a relative tolerance $|a - b| > 10^{-6} \max(1, |a|)$. Trends group flights by the profile's aircraft key and plot each metric in chronological order.

Plot down-sampling. Sending 10^6 points to a chart is wasteful, and naive decimation hides spikes. The API keeps both extremes per bucket. For n samples in an epoch and a budget of M points,

$$B = \max\!\big(1, \lceil n/(M/2) \rceil\big), \qquad \text{bucket}(i) = \Big\lfloor \frac{i-1}{B} \Big\rfloor,$$

and for each bucket it returns $\arg\min v$ and $\arg\max v$. Every spike survives in the plot. This envelope is used only for display and is never used for diagnosis.

12 3D replay

Positions are projected to a local tangent plane around the first fix (ϕ_0, λ_0) with the equirectangular approximation (error well below GPS noise over a few kilometres):

$$N = (\phi - \phi_0) \cdot 111\,320, \qquad E = (\lambda - \lambda_0) \cdot 111\,320 \cos\phi_0, \qquad U = h_{\text{home}}.$$

Because the coordinates are local metres, screenshots do not reveal the flying field. The aircraft glyph is drawn by rotating body-frame arm vectors (forward–right–down) to NED with

$$R_{\text{NED}\leftarrow\text{FRD}} = \begin{pmatrix} c_\psi c_\theta & c_\psi s_\theta s_\varphi - s_\psi c_\varphi & \cdot \\ s_\psi c_\theta & s_\psi s_\theta s_\varphi + c_\psi c_\varphi & \cdot \\ -s_\theta & c_\theta s_\varphi & \cdot \end{pmatrix},$$

using only the first two columns because arms lie in the body xy plane, and then converting to ENU $(E, N, -D)$ for plotting. The path is coloured by flight mode, with markers for takeoff, attitude excursion, crash report and waypoint status.

13 Flight simulator

The simulator exists to test the analyser against ground truth. It produces byte-exact `.tlog` captures using the real MAVLink encoder, so every parser, rule and plot runs on it unchanged. It is deterministic for a given seed.

13.1 Airframe

Mass $m = 1.6\,\mathrm{kg}$, arm $a = 0.18\,\mathrm{m}$, inertia $I = (0.022, 0.022, 0.04)\,\mathrm{kg\,m^2}$, hover throttle $h = 0.38$, yaw coefficient $k = 0.016$, linear drag $c_d = 0.35$. Maximum thrust per motor follows from hover:

$$T_{\max} = \frac{mg}{4h}, \qquad \ell = a \cdot \tfrac{\sqrt{2}}{2} \text{ (lever arm on X frame).}$$

13.2 Guidance

Toward target (n_t, e_t, u_t) at distance D:

$$v_{\mathrm{des}} = \min(5,\, 0.8D)\frac{(\Delta n, \Delta e)}{D}, \qquad v_{u,\mathrm{des}} = \mathrm{clip}\big(1.2(u_t - u),\, -1.5,\, 2\big), \qquad \psi_{\mathrm{des}} = \mathrm{atan2}(\Delta e, \Delta n).$$

A waypoint is reached when the horizontal distance is at most 2 m and the vertical distance below 2 m. RTL climbs to at least 10 m, returns and switches to LAND within 2 m of home.

13.3 Attitude and thrust control

Velocity error becomes acceleration demand $a_{n,e} = \mathrm{clip}(1.4(v_{\mathrm{des}} - v), \pm 4)$, rotated into the *estimated* heading $\hat{\psi} = \psi + \psi_{\mathrm{err}}$ (this is how a compass error corrupts the flight):

$$a_{\mathrm{fwd}} = a_n \cos\hat{\psi} + a_e \sin\hat{\psi}, \qquad a_{\mathrm{right}} = -a_n \sin\hat{\psi} + a_e \cos\hat{\psi},$$

$$\theta_d = \mathrm{clip}\big(-\mathrm{atan2}(a_{\mathrm{fwd}}, g), \pm 0.52\big), \qquad \varphi_d = \mathrm{clip}\big(\mathrm{atan2}(a_{\mathrm{right}}, g), \pm 0.52\big).$$

Collective thrust compensates tilt:

$$\tau_c = \frac{m\,(g + a_u)}{\max(0.3, \cos\varphi\cos\theta)\cdot 4T_{\max}}, \qquad a_u = \mathrm{clip}\big(2.5(v_{u,\mathrm{des}} - v_u), -4, 5\big).$$

PD attitude torques:

$$\tau_\varphi = I_x\big(100(\varphi_d - \varphi) - 16p\big), \quad \tau_\theta = I_y\big(100(\theta_d - \theta) - 16q\big), \quad \tau_\psi = I_z\big(9\,\mathrm{wrap}(\psi_d - \hat{\psi}) - 6r\big).$$

13.4 Mixer with ArduPilot-style priority

Per-motor roll/pitch and yaw contributions:

$$\rho_k = \frac{-x_k\tau_\varphi + y_k\tau_\theta}{4\ell T_{\max}}, \qquad \gamma_k = \frac{\sigma_k\tau_\psi}{4kT_{\max}}.$$

With limits $[s_{\min}, s_{\max}] = [0.15, 0.95]$:

1. Roll/pitch first: if $\max\rho - \min\rho > s_{\max} - s_{\min}$, scale ρ to fit.
2. Shift the collective so that $\tau_c + \rho_k$ fits the range: $\delta = \max(0, \max_k(\tau_c + \rho_k) - s_{\max}) - \max(0, s_{\min} - \min_k(\tau_c + \rho_k))$, $b_k = \tau_c - \delta + \rho_k$.
3. Yaw last, scaled by the remaining headroom: $\eta = \min_k \min(s_{\max} - b_k,\, b_k - s_{\min})$, $u_k = \mathrm{clip}\big(b_k + \mathrm{clip}(\eta / \max|\gamma|, 0, 1)\gamma_k\big)$.

This reproduces the real failure signature: when a motor weakens, yaw authority goes first, then roll/pitch saturate, then the attitude diverges.

13.5 Rigid-body dynamics

Motor forces $F_k = h_k T_{\max} u_k$ with health $h_k \in [0, 1]$ give

$$\tau_\varphi = -\ell \sum_k x_k F_k, \quad \tau_\theta = \ell \sum_k y_k F_k, \quad \tau_\psi = k \sum_k \sigma_k F_k,$$

$$\dot{p} = \tau_\varphi / I_x - 0.5p, \quad \dot{q} = \tau_\theta / I_y - 0.5q, \quad \dot{r} = \tau_\psi / I_z - 0.8r.$$

Translational acceleration from total thrust $A = \sum F_k / m$, rotated by attitude and with drag relative to the wind $\mathbf{w}(t) = (w_n + 0.6 \sin(t/7),\, w_e)$:

$$a_{\text{fwd}} = -A \sin\theta \cos\varphi, \quad a_{\text{right}} = A \sin\varphi, \quad a_{\text{up}} = A \cos\theta \cos\varphi - g,$$

$$\dot{v}_n = a_{\text{fwd}} \cos\psi - a_{\text{right}} \sin\psi - c_d(v_n - w_n), \quad \dot{v}_e = a_{\text{fwd}} \sin\psi + a_{\text{right}} \cos\psi - c_d(v_e - w_e), \quad \dot{v}_u = a_{\text{up}} - c_d v_u.$$

Integration is explicit Euler with $\Delta t = 10\,\text{ms}$. Ground contact zeroes velocity, and an impact faster than $2.5\,\text{m/s}$ is recorded.

13.6 Crash check

As in ArduCopter, in AUTO/RTL/LAND/LOITER the thrust-axis angle error ε is computed from the desired and actual attitude. If $\varepsilon > 30°$ for 2 s, the simulator emits `Crash: Disarming: AngErr=...` and disarms.

13.7 Fault models

Scenario	Model
`motor_failure`	Two seconds after WP2, motor 3 degrades linearly: $h_3(t) = \max\big(0.3,\, 1 - 0.06(t - t_f)\big)$.
`compass_interference`	Innovation ratio grows with throttle $\bar{u}$: $\rho_c = 0.04 + 6\max(0, \bar{u} - 0.2)$. The heading error integrates $\dot{\psi}_{\text{err}} \propto \max(0, \bar{u} - 0.25)$, clipped to $\pm 1.4\,\text{rad}$, so the aircraft toilet-bowls. After 4 s above 1 the EKF switches compass (error $\times 0.6$). After 25 s it triggers an EKF failsafe and LAND.
`vibration`	Vibration level $38 + 12\sin(t/3) + \mathcal{N}(0, 3)\,\text{m/s}^2$ on the z axis, random clip increments, and torque noise $\mathcal{N}(0, 0.01)\,\text{N m}$.
`radio_failsafe`	PreArm radio warnings. In flight after WP2 the RC channels drop to zero, followed by *Radio Failsafe* and RTL.

Battery: open-circuit voltage decreases by coulomb counting, $V_{oc} \mathrel{-}= I\,\Delta t / 3600 \cdot 0.9$, and the terminal voltage is $V = V_{oc} - 0.025 I$ with $I = 1.5 + 32\bar{u}\,\text{A}$. Logs are labelled `SIMULATED FLIGHT` and located at the public SITL field.

13.8 Validation

The test suite asserts the expected outcome of each scenario:

- *nominal*: no critical findings and every waypoint *reported_complete*;
- *motor_failure*: tracking loss, ceiling saturation and imbalance, with thrust/configuration ranked first;
- *compass_interference*: EKF ratio above 1 and a compass-selection event;
- *vibration*: persistent vibration and pre-event clipping.

A private regression capture of a real crash (ArduCopter 4.7.2, two AUTO crash-check disarms) is pinned by a JSON fixture.

14 The `.ant` analysis format

An `.ant` (*ANalysed Telemetry*) file is a finished, shareable analysis: a ZIP container, following the same convention as ODF and EPUB.

Entry	Contents
`mimetype`	`application/vnd.anatele.analysis+zip`; first entry, stored uncompressed
`meta.json`	format version, versions, original file hash, profile, per-entry SHA-256
`manifest.json`	the complete analysis
`observations.parquet`	decoded records (evidence)
`signals.parquet`	normalised and derived signals
`report.html`	printable report, readable without AnaTele
`original.<ext>`	optional; the untouched source log

Integrity. For each entry x, `meta.json` stores $H(x) = \text{SHA-256}(x)$, and the original log is bound by the hash that was computed at import. On opening, every entry is re-hashed while it is streamed and the file is rejected if $H'(x) \neq H(x)$. This detects accidental corruption and naive edits. It is *not* a signature, so it does not prove who created the file.

Untrusted-input handling. Only whitelisted entry names are accepted, so path traversal is impossible. Duplicate and encrypted entries are rejected, and only stored or deflate compression is allowed. Sizes are capped *during* decompression (which defeats zip bombs): 1 GiB per file, 768 MiB per Parquet, 1.5 GiB in total. JSON structure and the Parquet column schema are validated, and everything is extracted to a staging directory that is renamed into place atomically or deleted. Nothing in the file is executed.

Privacy. Without the original log the recipient still sees decoded records and the flight path, but no raw bytes and no re-analysis. Both variants contain GPS positions, and the UI says so before saving.

15 Security and distribution

- The web UI binds to `127.0.0.1` and has a Host allow-list. It rejects cross-origin writes and sends `nosniff`, `DENY` framing and `no-referrer` headers. All dynamic HTML is escaped.
- The worker runs as a separate process with time and size limits. It is isolation, not a sandbox.
- Releases are built on GitHub Actions with SHA-pinned actions and read-only build jobs. Only a separate publish job holds write and attestation permissions. Runtime dependencies are installed from a hash-locked requirements file (`-require-hashes`). Release assets come with `SHA256SUMS.txt` and signed SLSA build-provenance attestations.

16 Interfaces

Desktop. A PySide6 window in the style of openMotor:

- left: recordings, profile and episodes;
- centre: stacked matplotlib plots with event and phase markers;
- right: X/Y axis signal selection;
- bottom: findings, hypotheses, rules, events, parameters, missions, flight path, evidence and statistics.

Clicking any plotted point opens the exact record (decoded payload and raw bytes).

Web/API. FastAPI with the same analysis. Endpoints cover logs, events, rules, parameters, parameter diffs, series, positions, records, reports, `.ant` export and import, comparison and trends.

CLI. `anatele [web|simulate|replay|analyse|export]`.

17 Limitations

- Telemetry contains only what was streamed or recorded. Absence is never a healthy result.
- There are no calibrated probabilities, no per-core EKF innovation analysis from link telemetry, and no direct motor-failure diagnosis. Motor commands are not measured thrust.
- Phase inference without landed-state messages is heuristic and labelled as such.
- Clock segmentation is conservative. Capture time is not sensor time.
- Positions are estimates, and waypoint proximity is not proof of completion.
- The simulator is deliberately simple and proves nothing about a real aircraft.

18 Results and outlook

18.1 Our crash log

The capture is 2.2 MB and contains 53 465 records from ArduCopter 4.7.2 (`fb773567`). It has no battery monitor and no landed-state messages. AnaTele found the following:

- **Sources.** It separated the vehicle (1:1), the ground station (255:190) and the radio (51:68). It also decoded 973 parameters using the C-cast encoding of the semantic pack.
- **Episodes.** There were three armed episodes. In two of them the aircraft flew: takeoff was *inferred* at about 640 s and 980 s, and each flight ended in a crash-check disarm (`AngErr=160>30`) about 17–20 s after takeoff, just after mission item 2 had started. The third episode showed no airborne evidence.
- **Compass and EKF.** The EKF compass test ratio rose from about 0.02 on the ground to about 2 after takeoff in *both* flights. In the first flight a compass-selection event came 6.8 s before the disarm. Before arming there were repeated *Compasses inconsistent* PreArm messages. This all agrees with the calibration problem we found.
- **Motors.** Before the event, the CW pair (motors 3 and 4) was commanded consistently above the CCW pair (1 and 2). The mixer decomposition labels this a yaw-axis imbalance: the controller was fighting a heading error, which is not the same as a weak motor.
- **Vibration.** Vibration and clipping persisted before the event in the first flight but not in the second, so vibration alone does not explain both crashes.
- **Configuration.** `BATT_MONITOR=0` and `FS_GCS_ENABLE` had changed from 1 to 0 during the session. `ARMING_SKIPCHK` bit 7 was set, which skips the board-voltage check. Power was therefore reported as *unassessable*, not healthy.
- **Missions.** Waypoint 2 was geometry-consistent with completion in flight one (0.4 m). In flight two it was not visited in the observed trajectory (about 22 m).

A single-sentence answer of "compass" was right, but this is the detail it missed. Next steps include onboard DataFlash-specific rules (XKF4 per-core innovations, MAG field versus throttle

for compassmot, ESC RPM/current), FFT-based vibration analysis, signed `.ant` files and more frame geometries.

AnaTele is MIT-licensed. Findings are evidence-linked observations and screening hypotheses, never a crash verdict, probability or airworthiness approval.